

UPPSALA, Sweden

WMO: 024580

Lat: 59.897N

Lon: 17.589E

Elev: 68

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.2	3.9	-6.0	4.0	-1.7	0.3	5.6	4.5	24.1	35.2	21.6	33.1	2.0	310	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	80.9	65.8	77.4	63.9	74.2	62.3	67.8	76.6	65.7	74.3	63.9	71.4	8.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.5	71.2	62.6	85.0	68.9	60.8	79.7	67.2	32.1	76.6	30.5	74.6	29.2	71.3	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	18.7	16.5	-9.0	84.9	8.3	3.6	-15.0	87.5	-19.8	89.5	-24.5	91.5	-30.5	94.1
			-9.2	69.8	8.1	3.0	-15.0	72.0	-19.8	73.7	-24.4	75.4	-30.3	77.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	26.3	25.8	31.8	41.2	50.0	57.5	62.7	60.7	52.9	43.3	35.4	29.5
	DBStd	14.87	9.87	9.96	8.35	6.47	6.67	5.41	5.04	5.14	5.72	6.61	7.31	9.63
	HDD50	3675	736	678	563	273	85	6	0	1	36	223	438	636
	HDD65	8022	1201	1098	1028	714	467	231	108	152	363	671	888	1101
	CDD50	1188	0	0	0	9	83	232	393	332	123	16	0	0
	CDD65	63	0	0	0	0	1	7	36	18	1	0	0	0
	CDH74	724	0	0	0	0	20	97	428	176	3	0	0	0
	CDH80	126	0	0	0	0	1	8	97	20	0	0	0	0
	WSAvg	7.8	8.2	8.1	8.1	8.2	8.1	7.8	6.8	6.9	7.6	7.9	7.8	8.2
	PrecAvg	22.4	1.6	1.1	1.1	1.3	1.4	1.9	2.8	2.8	2.3	2.1	2.2	1.8
(15) Precipitation	PrecMax	30.1	3.1	2.9	2.2	3.7	4.1	4.6	5.8	7.2	5.7	4.9	4.9	4.4
	PrecMin	15.8	0.2	0.1	0.0	0.1	0.1	0.4	0.2	0.6	0.2	0.4	0.4	0.5
	PrecStd	3.6	0.7	0.6	0.6	0.7	0.9	1.0	1.4	1.6	1.3	1.1	1.1	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	45.8	55.9	67.8	76.5	81.0	87.0	82.5	73.1	60.9	52.8	49.8
		MCWB	42.3	42.2	46.9	53.9	60.7	64.9	67.7	65.7	61.3	55.6	50.4	47.3
	2%	DB	41.9	42.3	50.2	62.4	72.5	76.9	82.7	79.0	69.0	57.6	49.9	45.8
		MCWB	40.2	39.4	42.8	50.7	58.2	62.5	66.7	65.5	60.3	54.1	48.2	44.1
	5%	DB	39.8	40.1	45.9	58.0	67.8	73.2	79.0	75.3	65.8	55.5	47.6	43.2
		MCWB	38.3	37.8	40.7	48.5	55.6	60.6	65.1	63.9	58.6	52.8	46.1	41.7
	10%	DB	37.9	37.9	42.4	53.8	63.5	69.7	75.2	71.9	62.8	53.3	45.5	40.7
		MCWB	36.6	35.8	38.3	45.5	53.1	58.7	63.3	62.2	57.0	50.9	44.3	39.4
	0.4%	WB	43.1	42.8	47.0	55.3	62.6	66.8	71.0	70.4	64.1	56.9	50.9	47.5
		MCDB	44.2	44.7	54.5	65.5	73.9	77.2	81.6	78.2	70.2	59.4	51.9	49.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.4	40.0	43.7	51.7	59.5	63.8	68.6	67.3	61.3	54.8	48.3	44.2
		MCDB	41.4	41.8	49.0	60.5	69.0	73.7	78.4	74.7	66.8	56.9	49.4	45.3
	5%	WB	38.6	38.0	41.2	49.1	56.7	61.8	66.5	65.1	59.6	53.3	46.3	41.8
		MCDB	39.6	39.7	45.0	57.1	66.6	71.0	76.2	72.1	64.6	55.2	47.4	42.9
	10%	WB	36.7	36.2	39.0	46.4	54.1	59.9	64.7	63.4	57.8	51.2	44.4	39.5
		MCDB	37.7	37.6	41.9	52.5	61.8	67.7	73.0	69.8	61.9	52.9	45.4	40.4
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	11.2	13.4	17.4	18.8	17.6	18.8	18.1	15.5	11.2	7.9	8.5
		MCDBR	8.4	9.7	17.6	25.0	27.5	24.8	26.5	23.9	19.4	12.8	9.8	8.5
		MCWBR	7.8	8.1	11.7	15.1	15.1	12.7	12.3	12.0	11.6	9.9	9.0	7.9
	5% WB	MCDBR	8.2	8.8	16.3	23.2	24.1	21.5	22.3	19.7	17.0	11.4	9.4	8.1
		MCWBR	7.9	7.6	11.4	14.4	14.1	11.9	11.3	11.0	10.8	9.3	8.8	7.8
(40) Clear-Sky Solar Irradiance	taub		0.273	0.315	0.336	0.371	0.369	0.360	0.385	0.377	0.343	0.331	0.299	0.256
	taud		2.198	2.273	2.351	2.327	2.391	2.443	2.392	2.412	2.497	2.422	2.250	2.121
	Ebn at Noon		161	217	252	262	272	277	266	259	250	212	148	117
	Edh at Noon		13	22	27	33	34	32	33	30	24	19	13	10
(44) All-Sky Solar Radiation	RadAvg		93	294	770	1241	1605	1790	1650	1300	847	379	119	60
	RadStd		14	46	94	121	152	117	164	130	99	62	27	6

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	-1.84	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	-203	-261	N/A	N/A

Nomenclature: See separate page